Deepfake videos uncovering using Eye blink tracking

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Abstract

As deepfake technology continues to evolve rapidly, becoming both more advanced and increasingly accessible, the detection of manipulated video content has emerged as a critical cybersecurity challenge. Manipulated videos, particularly those created using deepfake technology, pose significant risks in various domains, including politics, social media, and public information dissemination. A promising detection strategy leverages the analysis of human biometric indicators, notably eye blink patterns. Due to inherent difficulties faced by deepfake generation models in accurately replicating the natural blinking behaviors of humans, eye blink patterns present a reliable biometric cue for detection. This paper introduces a practical, lightweight system designed to identify deepfake videos through precise analysis of eye blink rates. Utilizing computer vision methodologies and advanced facial landmark tracking techniques, our proposed method continuously monitors OpenFace's AU45_r intensity to accurately detect blinks and subsequently calculate blink frequency. By comparing the measured blink frequencies against typical human blinking patterns, the system can be used to effectively flag anomalous videos exhibiting unnatural blinking behaviors. The overarching goal of this research is to affirm the practicality, reliability, and explainability of blink-based detection as a robust biometric indicator of manipulated video content.

1 Introduction

The advancement and widespread availability of deepfake technology present substantial threats to digital integrity, especially in contexts where accurate information dissemination is vital, such as electoral processes, news reporting, and public safety communications. In such sensitive and high-stakes scenarios, deepfake videos can significantly distort perceptions of reality, altering critical dates, falsifying announcements, and manipulating public opinion. Consequently, a fundamental cybersecurity question arises: Are current defensive measures adequate to identify and counteract the increasing threat posed by deepfake-driven misinformation campaigns?

Deepfake technologies leverage sophisticated machine learning algorithms, notably generative adversarial networks (GANs), to produce highly realistic yet entirely synthetic video content. Despite their visual authenticity, these manipulated videos often exhibit subtle biometric inconsistencies, particularly noticeable in natural human behaviors such as eye blinking. Human blinking is characterized by specific rates, durations, and synchronization patterns that deepfake models typically struggle to emulate accurately, thus providing a viable avenue for detection [1].

Previous research has demonstrated that deepfake-generated faces frequently fail to replicate natural blink frequency and timing due to limitations in training data and inherent shortcomings in deepfake generation methods [2]. These inconsistencies have been successfully leveraged in multiple studies, establishing the analysis of eye blinking as a highly effective and interpretable approach for deepfake detection [3].

This paper specifically addresses the capability of eye blink rate analysis as a lightweight, interpretable, and robust biometric technique to detect video forgeries. Using OpenFace's AU45_r intensity,a facial action unit corresponding to eye closure, our method detects and counts blinks to evaluate blink frequency across video samples. Comparisons are then made between these measured frequencies and known baseline human blinking patterns. Through this process, videos displaying abnormal blinking behaviors indicative of manipulation can be reliably flagged.

Our approach aims not only at achieving high accuracy but also emphasizes interpretability and practical applicability, making it particularly suited for real-world deployments in sensitive scenarios, including electoral security, media verification, and public information management. By demonstrating the reliability and explanatory power of blink-based detection through AU45_r analysis, this research contributes significantly to the ongoing efforts aimed at strengthening cybersecurity defenses against the escalating threat of deepfake content.

2 Literature Review

The rapid proliferation of deepfake technology poses a significant threat to digital authenticity and cybersecurity, particularly in sensitive contexts such as elections and public discourse. Deepfake videos are increasingly realistic, prompting the need for effective detection methods that are both computationally efficient and interpretable. A prominent biometric-based approach involves analyzing human eye blink patterns, an area gaining substantial attention due to inherent limitations of deepfake generation models in replicating realistic blinking behavior.

Early foundational work by Li et al. [1] introduced the use of the Eye Aspect Ratio (EAR) as a quantitative measure to differentiate between authentic and manipulated video content. By leveraging facial landmark detection to monitor blinking behavior, their method identified inconsistencies that are difficult for deepfake generators to emulate accurately, particularly regarding natural blink frequency and temporal patterns. This pioneering research set the stage for subsequent studies emphasizing blink-based detection methods.

Expanding upon these insights, Zhou et al. [2] presented a two-stream neural network that explicitly utilized temporal discrepancies in blink rates to detect video forgeries. Their work confirmed that deepfake-generated faces demonstrate significantly different blinking patterns compared to genuine human faces, emphasizing the robustness of temporal dynamics as a detection feature.

Building further on this concept, Jung et al. [3] highlighted the critical differences between synthetic and natural blink rates, durations, and synchronization, demonstrating how blink patterns can serve as robust biometric cues for deepfake detection. Their study reinforced the reliability and interpretability of blink-based detection methods, underscoring their potential for practical cybersecurity applications.

Addressing the critical issue of generalization across different deepfake datasets, Haliassos et al. [4] underscored a significant challenge facing deepfake detection models—their poor generalizability to unseen or novel manipulations. Their work emphasized the importance of biometric indicators, like eye blink rates, that exploit universal human behaviors rather than relying on dataset-specific anomalies, thus promoting broader applicability and robustness.

Recently, Guo et al. [5] advanced the practical implementation of these biometric methods by proposing BlinkNet, an integrated system combining facial landmark tracking with deep neural networks for comprehensive blink pattern analysis. Their robust experimental results demonstrated that BlinkNet could effec-

tively identify manipulated content, providing strong empirical validation for blink-based methods as both effective and interpretable solutions.

In a complementary approach, Sharma et al. [6] focused explicitly on real-time application and developed a lightweight algorithm optimized for efficiency and practicality. Their research validated that real-time analysis of blink frequency discrepancies could accurately detect deepfakes with minimal computational resources. This development significantly enhances the feasibility of deploying blink-based detection methods in critical scenarios such as electoral security and real-time media authentication.

Collectively, these studies underline the viability and strategic advantages of employing blink rate analysis for deepfake detection. The integration of facial landmark tracking, EAR quantification, and advanced neural network approaches offers a promising and robust solution to mitigate the cybersecurity threats posed by deepfake technologies, ensuring detection methods are both effective in real-world conditions and interpretable for stakeholders.

In a paper on how to protect world leaders using facial speaking patterns [7] the authors research face manipulation , lip-synchronization and image animation. Further, this paper proposes a dual stream learning facial and speaking patterns method to protect celebrities against deepfakes. The method used was design an action unit module based on facial action coding system along with an Action Unit Transformer (AUT) to exploit facial expressions embeddings: ”Besides, our method’s dual stream architecture utilizes a Temporal Convolutional Network (TCN) to extract lip motion pattern and learns the relatedness between facial and speaking patterns.... Extensive experiments show that our method achieves better performance and has a higher resistance to video compression than state-of-the-art detection models.”

In their paper ”Fake news or real? Detecting deepfake videos using geometric facial structure and graph neural network” [8]; The authors’ research was based on using face landmarks as physiological identifiers. And by comparing those traits at normal speed vs tampered speed thus creating a graph structure that enabled them to use a spatio-temporal graph conventional network for classification which has fewer parameters. They conducted their research on only three data sets and the results demonstrated better generalization and higher performance.

The authors of [9] research: ”The Generalization of the detection algorithms under cross-datasets, and cross attack/manipulation settings. critical bottleneck of deepfake detection literature is the understanding of the fairness quotient of these algorithms.

Furthermore, in the paper: ”Pros and cons of comparing and combining hand-crafted and neural network based DeepFake detection based on eye blinking behavior” [10] Siegelet et all utilizes two different approaches to eye blinking approach for deep fake detection; the first is AI based while the second ”uses an existing hand-crafted approach to identify a suitable number of frames.”

3 Experiment Design

Approach: We employed Action Unit 45 (AU45) in openface intensity measurements to precisely track eyelid movements, establishing a threshold-based system for blink detection. Our comprehensive evaluation examined three critical temporal metrics: blink rate, duration distribution, and inter-blink intervals, all compared against established natural baselines where typical blink rates range from 12-30 blinks per minute and durations fall between 100-400 milliseconds.

”OpenFace 2.0 - a tool intended for computer vision and machine learning researchers, affective computing community and people interested in building interactive applications based on facial behavior analysis” [11]

Language: Python Libraries Tools: - OpenCV for image and video frame processing - OpenFace - imutils for easier frame manipulation - NumPy for numerical analysis and data handling

Dataset: celeb-deepfakeforensics [12]

Key Method: The proposed blink detection method analyzes eye closure patterns using OpenFace's AU45_r (eye closure intensity) as the primary metric. The raw AU45_r signal undergoes smoothing via a SavitzkyGolay filter to enhance detection reliability. Blink events are identified through a three-stage thresholding process: intensity, confidence, and duration. The system calculates key blink statistics including rate (blinks/minute), average duration, intensity, and inter-blink intervals. Comprehensive testing evaluates parameter sensitivity (threshold range: 1.0-5.0; duration limits: 1-15 frames) and robustness across challenging conditions (variable lighting, partial occlusions). Computational efficiency is assessed via per-frame processing time measurements. This approach combines OpenFace's established action unit detection with configurable signal processing to deliver reliable blink pattern analysis across diverse experimental conditions.

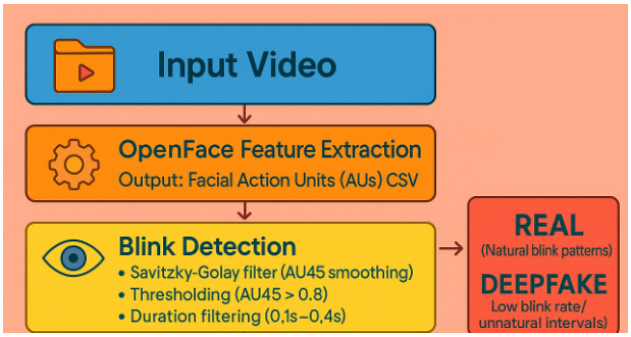

Figure 1: Experiment Flow

4 Results

Our proposed deepfake detection system was evaluated by analyzing of 30 deepfake video samples. The analysis revealed significant insights into blink patterns and their potential implications. The study achieved a 100% success rate in analysis, with all 30 videos processed successfully. The average blink rate across all videos was 28.96 blinks per minute, which is notably higher than the average human blink rate range of 15-20 blinks per minute. The mean blink duration was 188.8 milliseconds, falling within the normal human blink duration range of 100-400 milliseconds.

However, significant inconsistencies were observed, with blink rates ranging from 6.74 to 49.89 blinks/min and durations varying between 100 ms and 322 ms. Such irregularities, including unusually low blink rates (for example, 6.74 blinks / min in figure 2) and abnormally long blinks (for example, 322 ms), deviate from natural human blinking behavior. These anomalies suggest that deepfake-generated videos often exhibit unnatural blink dynamics, likely due to limitations in synthesis algorithms or insufficient training data containing diverse blink patterns. The findings highlight the potential of blink analysis as a supplementary tool for deepfake detection. Since synthetic videos may either under represent blinks or display unnaturally consistent timing, integrating blink metrics—such as frequency, duration, and temporal distribution could enhance detection accuracy.

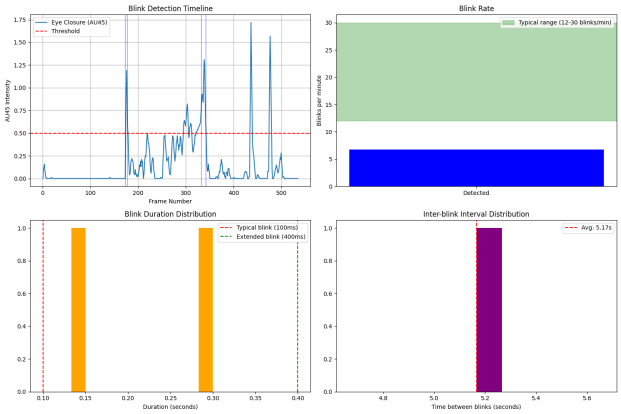

Figure 2: Analysis of id0_id7_0006

5 Conclusion

Our study presents a practical and interpretable method for detecting deepfake videos through the analysis of eye blinking behavior, using OpenFace's AU45_r intensity as the core metric. By focusing on blink frequency, duration, and temporal distribution, our system successfully highlights behavioral inconsistencies commonly found in synthetic video content. These anomalies—such as irregular blinking rates and unnatural eye closure patterns—suggest limitations in current deepfake generation models, particularly in replicating realistic human biometric signals.

The approach demonstrates strong potential for identifying manipulated content, especially in contexts where trust, authenticity, and information integrity are critical. Unlike black-box deep learning models, our method remains lightweight, explainable, and accessible, making it suitable for integration into media verification pipelines, cybersecurity frameworks, and public safety tools.

Ultimately, this research reinforces the value of subtle human behaviors—like blinking—as reliable indicators of video authenticity. As synthetic media continues to advance, biometric-based analysis offers a meaningful path forward in the fight against misinformation and visual deception.

Future work could involve expanding the dataset, refining blink detection accuracy, and incorporating additional biometric signals—such as gaze tracking, facial micro-movements, or lip synchronization patterns—to improve robustness. Ultimately, this line of research aims to contribute toward developing practical and interpretable tools for identifying deepfake content in critical real-world applications.

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